

Grid Vision T&D Sdn.Bhd.

Factory Acceptance Test Record DF1725IED RTU

Project Name	
Contract No.	

Approval				
	Name	Company	Date	Signature
Prepared by:	Muhammad Nor Farhan	Grid Vision T&D	17 th December 2024	
Reviewed by:	Lim Yee Chin	Grid Vision T&D	17 th December 2024	
Verified by:	Ir. Ts. Ramlah Binti Awang	Tenaga Nasional Berhad		

	
☒ NO FURTHER COMMENTS ☐ REVISE AND RESUBMIT ☐ INFORMATION ONLY	
By : 	Date : 19/12/24
This endorsement shall not : i. Be construed as acceptance of Variation and/or Contractor's Claim, and/or ii. Relieve the Contractor from any Contractual obligations or responsibilities	

FORM REF	1725IED-FAT-TR		
FORM DESCRIPTION	RTU FAT RECORD	FORM VERSION	02
PROJECT DETAILS	RTU SERIAL NO:	PRODUCT DETAILS	DF1725IED RTU
	CONTRACT NO:	TEST RECORD SHEET	1 of 39
FAT – TEST RECORD REF:	TABLE OF CONTENT	RECORD SHEET NO.	1 of 1

Change Log

No.	Date	Rev No.	Remark
1	22/12/2022	00	Initial Release
2	24/2/2024	00	Revise parameter setting for AI, IEC101.
3	16/5/2024	01	Separate Test Procedure and Test Record
4	10/9/2024	01	- Update firmware version and date for MCU-1-A and MCU-4-A. - Change Force to 0 setting under AI Card Module.
5	17/12/2024	02	- Add Ecolink LED in accessories list. - Add another firmware check under Processor Functional Test. - Update IEC104 Channel Redundancy Test Procedure. - Combine Physical and Functional result in 1 page. - Add checkbox and signature box in FAT flowchart. - Add tester's competency verification. - Update IOA for I/O Fail. - Add COM-6-A Module. - Add Appendix for I/O Test Setup.



FORM REF	1725IED-FAT-TR		
FORM DESCRIPTION	RTU FAT RECORD	FORM VERSION	02
PROJECT DETAILS	RTU SERIAL NO:	PRODUCT DETAILS	DF1725IED RTU
	CONTRACT NO:	TEST RECORD SHEET	2 of 39
FAT – TEST RECORD REF:	TABLE OF CONTENT	RECORD SHEET NO.	1 of 1

Table of Contents

Overall FAT Result.....	5
Factory Acceptance Test Summary	6
FAT Log.....	6
FAT Attendance.....	6
Pre-Requisites	
1.0 Approved Engineering Drawings	7
2.0 Bill of Quantity	7
3.0 SGP Certificate	8
4.0 RTU Panel IP Certificate	8
5.0 Test Equipment Record.....	8
6.0 Measuring Equipment Record	8
7.0 Software Record.....	9
8.0 Pre-FAT Record.....	9
9.0 Tester's Competency	9
Declaration	
1.0 Product Declaration	10
2.0 Test Setup.....	11
Accessories Declaration and Inspection	
1.0 Electronic Accessories	12
2.0 RTU Panel Accessories.....	13 - 15
Inspection – RTU Panel	
1.0 Panel Information.....	16
2.0 Physical and Quality Inspection.....	16
Inspection – Subrack	
1.0 Subrack Information	17
2.0 Physical and Quality Inspection.....	17
Inspection – Power Supply Module	
1.0 Physical and Quality Inspection	18
2.0 Functional Test – Power Supply	18

FORM REF	1725IED-FAT-TR		
FORM DESCRIPTION	RTU FAT RECORD	FORM VERSION	02
PROJECT DETAILS	RTU SERIAL NO:	PRODUCT DETAILS	DF1725IED RTU
	CONTRACT NO:	TEST RECORD SHEET	3 of 39
FAT – TEST RECORD REF:	TABLE OF CONTENT	RECORD SHEET NO.	1 of 1

Inspection – Processor

1.0 Physical and Quality Inspection	19
2.0 Functional Test - Processor	19
3.0 IEC101 Initialization	20
4.0 IEC104 Initialization	20
5.0 DNP3 Initialization	21

Inspection – COM-6-A

1.0 Physical and Quality Inspection	22
2.0 Functional Test – COM-6-A	22

Inspection – Digital Input Module

1.0 Physical and Quality Inspection	23
2.0 Functional Test – Digital Input Module	23

Inspection – Digital Output Module

1.0 Physical and Quality Inspection	24
2.0 Functional Test – Digital Output Module	24
3.0 Dummy Breaker and CES Functional Test.....	25
4.0 Buzzer Test	25

Inspection – Analog Input Module

1.0 Physical and Quality Inspection	26
2.0 Functional Test – Analog Input Module	26

Inspection – Analog Output Module

1.0 Physical and Quality Inspection	27
2.0 Functional Test – Analog Output Module	27

RTU Power Up 28

Parameter Setting

1.0 Processor Parameter Setting	
1.1 MCU-1-A.....	29
1.2 MCU-4-A.....	29
2.0 Digital Input Parameter Setting	
2.1 Under Card Module	30
2.2 Under Logical Module.....	30

FORM REF	1725IED-FAT-TR		
FORM DESCRIPTION	RTU FAT RECORD	FORM VERSION	02
PROJECT DETAILS	RTU SERIAL NO:	PRODUCT DETAILS	DF1725IED RTU
	CONTRACT NO:	TEST RECORD SHEET	4 of 39
FAT – TEST RECORD REF:	TABLE OF CONTENT	RECORD SHEET NO.	1 of 1

3.0 Digital Output Parameter Setting	
3.1 Under Card Module	31
3.2 Under Logical Module.....	31
4.0 Analog Input Parameter Setting	
4.1 Under Card Module	32
4.2 Under Logical Module.....	32
5.0 Analog Output Parameter Setting	
5.1 Under Card Module	33
5.2 Under Logical Module.....	33
6.0 IEC101 Parameter Setting.....	34
7.0 IEC104 Parameter Setting.....	35
8.0 DNP3 Parameter Setting	36
Virtual Alarm Test	37
Channel Redundancy Test	38
1.0 IEC101	38
2.0 IEC104	38
3.0 DNP3	38
Limit of Authority.....	39



FORM REF	1725IED-FAT-TR-TS		
FORM DESCRIPTION	RTU FAT RECORD - TEST SUMMARY	FORM VERSION	02
PROJECT DETAILS	RTU SERIAL NO:	PRODUCT DETAILS	DF1725IED RTU
	CONTRACT NO:	TEST RECORD SHEET	5 of 39
FAT – TEST RECORD REF:	OVERALL TEST RESULT	RECORD SHEET NO.	1 of 1

1.0 Overall Pre-FAT/FAT Result

RTU Serial No.	
----------------	--

PASS ☒

FAIL ☐

REMARKS:



Tested By:
(Signature/Official Stamp)



Tester's
Name:
Date:

Witnessed By:
(Signature/Official Stamp)



TNB QAI's
Name: Muhammad Azamuddin Bin Hasan
Date: